Preliminary Progress Review (PPR) Report of 25_26J_009

Project ID: 25_26J_009

Project Title: A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

Title of the Individual Component: Lightweight IDS for Simultaneous Detection of Sinkhole & Selective Forwarding Attacks in RPL

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1. Introduction

1.1. Purpose

This Preliminary Progress Review (PPR) document is intended to be the first summary of the progress of the confirmed research project, describe the research pathway, and justify the credibility and viability of the endeavor according to the module requirements for the Research Projects. This document serves as a review to the academically interim report which captures the essence of the problem, the gaps in the existing body of literature, the questions and the objectives of the research, the methodology, the outcomes, as well as the steps towards the implementation of the project at this point in time in the life cycle of the project.

Supervisors, co supervisors, members of the evaluation panel, and the relevant academic reviewers, who will be assessing the progress of the research project, constitute the audience of the PPR. The document is aimed at making evaluation easier, and describes the completed work in a structured, logical, and document-supported manner. It also describes the strategy to be followed in the coming phases of research.

1.2 Definitions, Acronyms, and Abbreviations

Acronym / Term	**Definition / Description**
IoT	Internet of Things
LLN	Low-Power and Lossy Network
RPL	IPv6 Routing Protocol for Low-Power and Lossy Networks
DODAG	Destination-Oriented Directed Acyclic Graph
IDS	Intrusion Detection System
CDFR	Control-to-Data Flow Ratio
PTI	Packet Time Interval
SIS	Sequence Integrity Score
UDP	User Datagram Protocol
DIO	DODAG Information Object
DAO	Destination Advertisement Object
DIS	DODAG Information Solicitation
PPR	Preliminary Progress Review
WBS	Work Breakdown Structure
Cooja	Network simulator used for Contiki-NG
Contiki-NG	Open-source operating system for IoT devices
Root Node	The RPL node that initializes and manages the DODAG
Observation Window	Fixed time interval over which IDS metrics are computed

Acronym / Term	Definition / Description
Severity Score	Aggregated risk value derived from multiple IDS metrics

1.3 Overview

The format of this Preliminary Progress Review (PPR) is designed to facilitate the review of the research project in its present state. Following the introduction, the rest of the document provides an extensive literature review, which builds the foundation of the research field and articulates the problem in the research field, based on the available scholarly resources. After this, the document dedicates one section to explain and justify the importance of the research problem, and to state the reason for the work to be carried out.

Thereafter, the research questions and the objectives of the research are stated so as to outline the focus and the expected results of the research. The next section of the document outlines the research methodology proposed to be used along with the approach, the conceptual framework, and the variables to be used in relation to the objectives stated. Lastly, the document outlines the sources of test data, the methods of data analysis, and the procedures to be followed to validate the research outcomes.

The final parts of the document describe the possible value of the research, state the boundaries and expected outcomes, and describe the limitations that may affect the process of carrying out the research. In order to construct the research outline, a project plan has been provided. The document ends with references and appendices to support and supplement the proposed research.

2. Literature review

Due to the reliance of RPL on cooperative behavior among resource constrained nodes, the trust based routing assumptions, and the absence of an enforcement means for RPL's intrinsic security, routing security in RPL based Internet of Things (IoT) networks is quite challenging. RPL is often used in low-power and lossy networks (LLNs) for environmental monitoring, smart cities, and disaster management. However, the absence of trust in the routing assumptions and the absence of enforcement means for intrinsic security, makes RPL open to numerous routing-layer attacks, more specifically sinkhole and selective forwarding attacks. These attacks impact the reliability of data delivery and overall stability of the network.

Shiranzadei's NADSA is one of the studies that addresses sinkhole attacks specifically. NADSA analyzes the routing control messages with the goal of determining the existence of sinkholes [1]. Although NADSA is effective in identifying sinkhole attacks, it is also effective only in the routing manipulation space and ignores the whole area of packet forwarding. This omission means that selective forwarding attacks, where an adversarial node routing packet(s) in the appropriate direction, drops or delays the packets at some point, go unnoticed.

Jiang and Liu focused on creating a trust-based defense mechanism to lessen the impact of selective forwarding attacks on RPL networks [2]. They assess a node's level of trust based on their packet forwarding activity, which helps identify the malicious dropping of packets. Still, this method is quite limited as it only targets packet forwarding, and neglects other important areas, such as sinkhole attacks. This illustrates the existing trust-based defense's biggest drawback, which is that the detection frameworks are compartmentalized to solve a single problem and neglect other areas, fail to address combined, coordinated, or synergistic attacks, and overlook the possibility of multiple sequenced attacks.

Research on intrusion detection that uses machine learning techniques to secure RPL-based networks is also beginning to emerge. Data-driven techniques to mitigate selective forwarding attacks have been studied by Biradar and Vishanath [3]. While this approach shows potential to achieve a considerable level of accuracy in the detection process, it comes at a significant cost in terms of the processing and memory needed to balance and address the challenges and limitations of the system. This is a significant concern as the system is a constrained system, operating on firewalled networks. In the real world, when deploying RPL networks, a large number of functionally constrained IoT motes apply strict limitations on energy and processing creating a bottleneck in the system and restricting the practical use of the approach deployed.

Routing enhancements, enabled through trust, have been suggested to increase resilience to routing attacks. Arshad et al. proposed THC-RPL, which is a lightweight trust-enabled routing protocol that attempts to provide a defense against Sybil attacks in RPL networks [4]. THC-RPL, while integrating trust in routing decisions and improving the resistance to some forms of identity attacks, is still limited to a few attack vectors and leaves combined sinkhole and selective forwarding attacks unaddressed. Additionally, the trust model does not have any real-time attack detection mechanisms to account for intrusions.

Under simultaneous sinkhole and selective forwarding attacks, Patel and Shah performed simulations and experiments centered around RPLs performance [5]. In these cases, while the performance of the network is analyzed, there is little to no discussion of lightweight mechanisms or detection frameworks to the vulnerabilities already identified.

Current research shows that although defensive strategies have been developed for individual sinkhole and selective forwarding attacks, the absence of lightweight, integrated intrusion

detection systems tailored for routing and forwarding attacks in RPL networks remains a significant gap. Most existing solutions have any one of the following shortcomings; excessive computational cost, restricted scope of attack detection, a centralized detection paradigm, or the absence of time-critical detection in lightweight IoT systems. Furthermore, many suggested solutions have yet to be validated in simulations that replicate the actual conditions of RPL networks.

This tells us that there is an absence of research into the development of lightweight, distributed systems for intrusion detection that have the ability to simultaneously and independently examine routing and packet forwarding to detect sinkhole and selective forwarding attacks. Closing this gap is crucial to enhancing the effectiveness of these systems.

2.1. Identification and significance of the problem

Due to their reliance on cooperative behavior and the lack of built-in granular security mechanisms, the RPL-based IoT networks are susceptible to numerous routing layer attacks. In regard to this, sinkhole and selective forwarding attacks can be considered one of the most serious routing-related threats due to their substantial negative impacts on the routing and delivery of data. Manipulation of the routing metric to attract data traffic via a sinkhole is one of the attacks, and selective forwarding is where he/she is in a position to drop or delay packets. In real-world situations, these two attacks are often combined, and a synergistic effect leads to severely negative impacts on the performance of the network.

Reviews of the literature indicate that most security initiatives are focused on the compartmentalization of these impacts. Some attempts to analyze routing data for inconsistencies center on the sinkhole and ignore packet forwarding behavior. In contrast, most of selective forwarding in terms of trust and anomaly models where the routing manipulation is ignored, while they focus on dropped packets. Therefore, the oversimplified view held in the literature leaves networks vulnerable when routing and forwarding behavior are exploited in concert. In addition, even though machine learning approaches to intrusion detection are relatively effective when the intrusion behavior is somewhat controlled, the limited capacity of the IoT mote leads to excessive uses of its computational and storage resources. This makes the solutions proposed in the literature impractical.

Another important constraint noted in the previous studies is the absence of lightweight and distributed intrusion detection mechanisms that can be deployed directly on the individual IoT nodes. Most of the existing solutions rely on centralized data monitoring, complex models of trust propagation, and/or computationally heavy, energy consuming cryptographic operations. This not only increases energy consumption but also shortens the lifetime of the network. In addition, the majority of studies defend their solutions against threats in a fixed and unchanging

network topology, and do not consider the effects of topology changes and node mobility. These are common in most of the real-world IoT deployments, and they strongly influence the behavior of the RPL protocol.

The gap in the literature is defined by the absence of a lightweight, resource-aware approach to intrusion detection that can be described as being able to simultaneously identify and classify sinkhole and selective forwarding attacks in RPL-based IoT networks. There is a need for a detection mechanism that observes and analyzes the routing control actions, monitoring the packet forwarding, and ensuring the data delivering intact, all at a low processing cost, in a fully distributed manner.The research in question demonstrates novelty in proposing a lightweight intrusion detection system that is capable of concurrent detection of sinkhole and selective forwarding attacks, as opposed to treating them as separate and independent attacks. The framework utilizes a combination of routing, dominance, timing, and integrity metrics to capture the various different dimensions of packet delivery, and aims to provide a consolidated solution for detection in low constrained environments in IoT. Unlike the current state of the art providing reliance on machine learning and central control systems, the proposed solution is a truly decentralized system, making it applicable in practice for real-world deployments of RPL.

From the scientific and technical aspects, the proposed system addresses an important problem of identifying the most critical gaps between the level of generated competence of the system while utilizing minimal resources in the IoT ecosystem. The ability to detect coordinated routing and forwarding attacks without increasing system overhead is a positive addition to the evolving state of secure routing for resource-constrained networks. The proposed solution increases the resilience of RPL-based IoT systems in critical use cases, such as disaster monitoring and smart infrastructure, where timely and reliable data delivery is of utmost importance.

3. Research Question

Considering the stated challenges in current RPL security mechanisms and a need for the lightweight and unified intrusion detection way, the following questions directs this research.

Q1: In RPL-based IoT networks, how can a lightweight intrusion detection system be developed to detect simultaneously sinkhole and selective forwarding attacks, without added significant computational and energy cost?

Q2: In resource-constrained RPL environments, when routing-level and forwarding-level behavioral metrics are combined, which metrics best constrain the detection of attacks, specifically sinkhole and selective forwarding, towards RPL?

Q3: Under both static and mobile RPL network scenarios, how well are dual routing and forwarding attacks detected by a distributed, threshold-based intrusion detection system?

Q4: What are the effects of the proposed lightweight intrusion detection system on network performance in terms of the packet delivery ratio, latency, and scalability when it is evaluated on Contiki-NG and Cooja?

4. Research Objectives

The main goal of this study is to develop and analyze a lightweight and distributed intrusion detection system that can identify and analyze sinkhole and selective forwarding attacks simultaneously, and that also keeps computational and energy costs low to suit resource-constrained devices.

The described goal can be broken down into several sub-objectives:

To study and analyze the behaviors of control packets and data packets in RPL-based IoT networks to discern behavior patterns attributed to sinkhole and selective forwarding attacks.

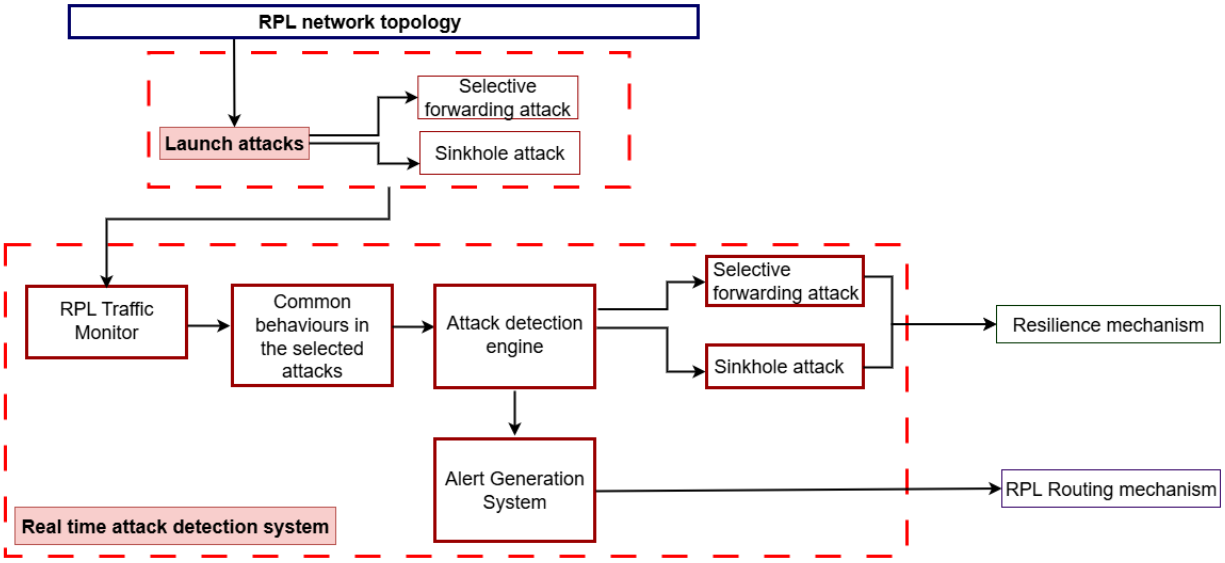
To develop a lightweight, threshold-based intrusion detection system that simultaneously detects sinkhole and selective forwarding attacks by monitoring the routing and forwarding layers of the network.

To realize the designed intrusion detection system in a distributed manner, at the level of individual nodes, while also being designed for resource-constrained IoT devices and steering away from centralized detection and machine-learning-based methods.

To analyze the proposed intrusion detection system in the context of the RPL network with and without mobility, with a focus on detecting the dual attacks in dynamic conditions.

To analyze the performance of the proposed intrusion detection system in terms of packet delivery ratio, latency, and scalability, using the Cooja simulator and Contiki-NG, among other networking parameters.

5. Research methodology



This research uses simulation-based experimental methods to design, implement, and assess a lightweight intrusion detection system (IDS) for RPL-based IoT networks. The choice of simulation-driven methods is because of its advantages in controlled experimentation, repeatability, and cost-effectiveness in assessing the security of routing mechanisms in resource constrained environments. Unlike other methodologies, simulation environments are used in myriads of RPL security and intrusion detection studies to assess, for example, attacks and detection efficiency [1], [4], [5].

This methodology comprises four distinct phases: baseline network modeling, design and implementation of the IDS, execution of the attack scenarios, and evaluation of the system's performance.

Modeling the Baseline Network:

Working RPL-based IoT networks are created using the Contiki-NG operating system along with the Cooja simulator which offer realistic emulation of the RPL behavior on constrained devices. These tools have historically been used for assessing routing attacks and other security processes in low power and lossy networks [4], [5]. A baseline scenario is created with the emulation of normal (attack-free) network function in order capture the standard behavior of routing control, packet forwarding and the patterns of data delivery. The observations from this baseline scenario are used to compute normal behavioral thresholds required for lightweight anomaly detection.

IDS Design and Implementation:

Out of design simplicity and cost concerns, the intrusion detection system (IDS) is developed as fully distributed, as opposed to incorporating a centralized architecture, that does any monitoring or machine learning. This would keep the computational and energy costs lower. Most of the RPL security studies to date have recommended distributed and lightweight detection because it is more viable and cost efficient for use with IoT devices [1],[2]. Especially for RPL, research suggests that simple statistical approaches work (or threshold detection, as it is called) with good effect [3].

The IDS monitors both routing and forwarding behaviors on the RPL network by using multiple metrics that work together to provide a complete picture of system performance. Each of these metrics is calculated independently on each network node and falls within a designated observation period, and each is compared to the detection thresholds that are established based on baseline simulations.

Feature Selection and Detection Logic:

The feature selection process is guided by conflict between the need to achieve effective attack detection and the need to keep energy and computation costs low. Instead of learning-based or high-dimensional features, a limited number of lightweight features are chosen that are directly linked to the behaviors of the network that are believed to cause sinkhole and selective forwarding attacks as documented in the previous research studies on RPL security [1],[2],[3].

The described features are obtained from observable network interactions. Each one is represented against the network protocol aspect the features represent:

1.Routing behavior feature:

Control-to-Data Flow Ratio (CDFR) is calculated based on the number of RPL control packets generated in relation to the amount of data packets forwarded in a defined time

frame. Sinkhole attacks adjust the control messages to draw in the traffic, in turn causing increases in routing control [1], [5].

2.Timing behavior feature:

PTI (Packet Time Interval) is calculated by observing the time gaps between packets. Selective forwarding attacks can produce unnatural delays that are caused by deliberately holding packets, or, by dropping some packets in the flow which is why the PTI can be an effective measure of forwarding behavior anomalies [2], [3].

3.Delivery integrity feature:

The Score of Sequence Integrity (SIS) is obtained by observing the patterns of missing packets and the disorder in the delivery that are usually described as the forwarding behavior of RPL networks [2], [3].

To minimize the processing effort, the features are computed individually in the nodes with the help of basic counters and timers. During each observation period, the values of the extracted features are evaluated against the thresholds from empirical data obtained from the behavior of an un-attacked network.

The detection logic functions using predetermined rules. When a feature exceeds the respective threshold, a binary anomaly flag is triggered. These anomaly flags are summed up to derive a consolidated severity score, indicating the confidence level of the malicious activity at a given node. The IDS, through multi-metric aggregation, successfully differentiates between discrete anomalies, coupled sinkhole attacks, and selective forwarding attacks, thus enabling the detection of multiple attacks in a decentralized manner, as evidenced in the work of [5].

Running Attack Scenarios:

In the simulated RPL network, sinkhole attacks and selective forwarding attacks are introduced, and both attacks are executed simultaneously to assess the effectiveness of the detection. Previous works suggested evaluating RPL security mechanisms focused on the combined attacks because of the synergistic effect of homogeneous inflow routing and selective forwarding attacks on the decline of the network performance [5]. The static network topology and mobility-enabled scenarios are used to observe the IDS response to evaluate the robustness of the dynamic routing.

Performance Evaluation and Validation:

The proposed IDS is evaluated on its effectiveness using standard network and detection metrics which include the packet delivery ratio, detection accuracy, false positive rate, and scalability. These metrics are used in RPL security evaluations, and aid in assessing the evaluation in terms of the security effectiveness and performance impact on the network [4], [5]. The experiments are carried out in the Contiki-NG and Cooja simulation tools for realistic evaluation and reproducibility under the constrained conditions of IoT devices.

6. Sources for test data and analysis

Test Data Sources:

Network data from simulations is the main source of test data for this study. It is the result of controlled experiments conducted with the Cooja network simulator and the Contiki-NG Operating System. These tools offer evaluatory environments for RPL-based Internet-of-

Things (IoT) networks and constrained devices. They have been used in much of the prior research in the security of RPL networks [4], [5].

Simulation scenarios have been set up for both normal (attack-free) and adversarial RPL networks. The test data from the experiments consists of:

- Normal RPL operation
- Sinkhole attack
- Selective forwarding attack
- Sinkhole and selective forwarding attack

These scenarios have been performed in unchanging network topologies and in topologies where node movement is permitted. This is to ensure that the experiments capture a wide variety of routing behaviors and dynamics of packet delivery.

The data collected from each test included:

- Count of control packets in RPL
- Logs of data packet forwarding
- Timestamps of packets sent
- Sequence numbers of packets
- Forwarding and routing statistics from each node

The data listed above aids the study in the analysis of intrusion detection focusing on the dominant routing, anomalies in timestamps, and the integrity of data delivery.

Process of Collecting Data :

While simulating, the process of data collection is fully automated in a repeatable manner. Each IoT node registers locally pertinent statistics of routing and forwarding using lightweight, environment specific counters and timers within Contiki-NG. Data collection occurs across predetermined observation windows in order to provide temporal consistency across the different experiments.

For each simulation run:

1. During standard operations of the network, node-level metrics are recorded to create baseline behavior.
2. During the attack scenario, the same metrics are captured in order to analyze anomalous behavior.
3. Multiple simulation runs are carried out in order to obtain and enhance the reliability of the results by eliminating randomness.

The simulation Cooja logs are generated, exported and processed to obtain feature reframing values related to the selected metrics of intrusion detection. This ensures the evaluation is reproducible and controlled, as well as eliminating the need to deploy in the real world at the early stage of the research.

Data Analysis Techniques:

Mathematical-statistical methods and rules-based methods that analyze lightweight intrusion data are applied to analyze collected data. Simulation logs are used to compute feature values for the following intrusion detection metrics:

- Control-to-Data Flow Ratio (CDFR)
- Packet Time Interval (PTI)
- Sequence Integrity Score (SIS)

Empirical thresholds are defined based on baseline feature distributions from the attack-free simulations. During attack scenarios, the feature values are evaluated against the predefined thresholds to detect the occurrence of abnormal activities. The occurrence/absence of threshold violations may be binary encoded, and the values are summed to a single. Anomaly status score to indicate the extent of confidence of the presence of a malicious activity.

The proposed intrusion detection system is assessed based on the quality, quantity, and extent of parameters based on the following standard recognition and detection metrics:

- Detection effectiveness
- Rate of falsely identified incidents
- Packet delivery effectiveness
- Change in the scalable and stable characteristics of the network

These metrics are standard in intrusion detection and RPL security studies and provide a metric for the objective evaluation of “normal” and “attack” scenarios [4], [5]. The analysis results are used to prove that the proposed system is capable of detecting sinkhole and selective forwarding attacks with minimal overhead.

Validation Strategy:

Engaging in the scenario involving the intelligences of systems designer (attack scenario) in the specified networks (normal scenario) will involve the validation of the proposed approach by checking the IDS responses in normal and attack scenario. IDS responses will involve normal routing behavior and malicious routing behavior. Successful detection will depend on the apprehension of:

- normal behavior and malicious behavior

- roaming coordinated routing and forwarding attacks
- complete detection (normal routing behavior) within the desirable network

Collecting and analyzing the data in the above described manner will ensure that the result of the research is validated empirically, reproducibly, and is within the scope of the research objectives.

7. Anticipated benefits

User Advantages:

The proposed lightweight intrusion detection systems offers users on RPL-based IoT networks practical and tangible advantages, especially in deployments that require high reliability, energy efficiency, and timely data delivery. The proposed system strengthens trust in the routing and data forwarding processes by simultaneously detecting sinkhole and selective forwarding attacks without a significant increase in the involved processing and communication costs.

Because of the system's IDS lightweight design, it localizes the detection of routing and forwarding anomalies, which enables rapid detection of affected rogue nodes and the reduction of compromised data. This is particularly relevant in mission-critical IoT scenarios, as in the case of disaster monitoring, environmental sensing, and smart infrastructure, where the absence or delay of certain sensor data can lead to incorrect actions or failure to respond to critical real-world situations.

Additionally, the distributed, on-device design of the IDS lightweight system means that minimal extra processing and energy costs will be placed on the IoT motes. By not employing centralized monitoring and machine-learning models, the proposed solution helps to preserve the system's overall longevity and scalability, making it ideal for sustained and extensive IoT deployments. In a broad sense, the IDS lightweight system improves the operational constraints flexibility for the end-users. It also balances the service providers expectations, therefore, enabling greater system resilience.

Contribution to the Body of Knowledge:

The presence of a gap in RPL intrusion detection has been addressed in the body of knowledge pertaining to the security of the IoT. While the majority of previous studies focus on sinkhole or selective forwarding attacks, this work targets the dual detection of both, proposing a unified, integrated, and lightweight solution. It provides the first example of a single detection attributed to a conjunction of both RPL routing and forwarding coordinated attacks, rather than relying on the traditional siloed approaches.

The research also pushes the boundaries of knowledge by showing the intrusion detection capability of a resource constrained environment through the integrated use of low behavioral overhead metrics, combined, in the multi-layered, threshold, and severity score schema. In contrast to a machine learning solution that is computationally intense, the proposed solution offers a means of detection that uses a sustainable processor and energy balanced resource and provides the necessary system processing.

The proposed intrusion detection system (IDS) has been implemented and evaluated using Contiki-NG and the Cooja simulator, which contributes empirical evidence of the feasibility and performance of lightweight, distributed intrusion detection in RPL (Routing Protocol for Low Power and Lossy Networks) networks, both under static conditions and with mobility enabled. These findings can be a resource for future research pertaining to the routing security, trust management, and intrusion detection in low-power IoT networks, as well as improve RPL-based systems.

8. Scope and specified deliverables / expected research outcome

Scope of Research:

This research focuses on the design, development, and assessment of a lightweight intrusion detection system (IDS) for RPL-based IoT networks. More specifically, this study focuses on the detection of sinkhole and selective forwarding attacks, both individually and in combination.

The proposed software solution functions at the routing and forwarding layers of the RPL protocol and is intended to deploy in a distributed fashion on IoT nodes with limited resources. The IDS is designed to monitor network behavior within its locality, compute simple metrics, and evaluate network behavior for anomalies based on thresholds without central analysis, and machine learning.

The system will accomplish the following:

- Monitor behavior of control routing and data forwarding in RPL networks
- Compute simple metrics on routing, packet timing and, delivery, and loss
- Detect sinkhole and selective forwarding attacks via an integrated mechanism of severity-based detection
- Function under both static and dynamic network conditions
- Be implemented and assessed using the Contiki-NG operating system and Cooja simulator.

Out of Scope:

In the interest of maintaining focus and feasibility, the following elements will be excluded from the scope of this research:

- Other than sinkhole and selective forwarding attacks, detection of other attack types (Sybil, wormhole, rank manipulation, etc.)
- The use of machine learning or deep learning models for intrusion detection
- The deployment of real-world hardware on physical IoT devices
- The implementation of automated node isolation, attack mitigation, or route repair mechanisms
- The integration of application layer security or the management of cryptographic keys

These boundaries allow the research to concentrate on lightweight detection, rather than network prevention or defense on a broader scale.

Specified Deliverables and Expected Research Outcomes

The anticipated deliverables for this research are:

1. A design for a distributed lightweight intrusion detection system for IoT networks based on RPL, which will allow for the simultaneous detection of sinkhole and selective forwarding attacks.
2. An IDS prototype that has been implemented within the Contiki-NG environment and integrated with the RPL protocol stack, designed for operation on resource-constrained IoT motes within the simulation.
3. A lightweight detection thresholds and metrics framework created from experimental validation and baseline network behavior.
4. Results from experimental evaluations that demonstrate the effectiveness of the IDS in normal, attack, and combined attack scenarios, as well as a variety of static and mobile network conditions.
5. Reports on performance relating to detection accuracy, in addition to the metrics on the false positive rate, packet delivery ratio, and the impact on scalability.
6. Technical documentation and research results pertaining to the system design, detection logic, experimental design, and evaluation results.

This research involves building a validated and resource-efficient intrusion detection system aimed to improve the security and reliability of RPL-based IoT networks while maintaining the operational constraints of low-power devices. The results are to be framed in a way to show that effective multi-attack detection can be accomplished while also maintaining optimal performance and scalability of the networks.

9. Research constraints (if any)

This study has some touchable boundaries that will dictate how the research will be carried out. These boundaries will not be acknowledged for the sake of neglecting the proposed work's feasibility or validity.

One such boundary is the dependence on simulated evaluation over deploying the evaluation on a physical hardware. Though the Contiki-NG OS and the Cooja simulator offer an abundant and realistic platform to evaluate RPL behaviour, simulated items are prone to a myriad of real-world hardware specific items like energy consuption variations, radio interference and noise. However, simulation is more then justified for the preliminary research stage, where the voulume of variables is typically excessive.

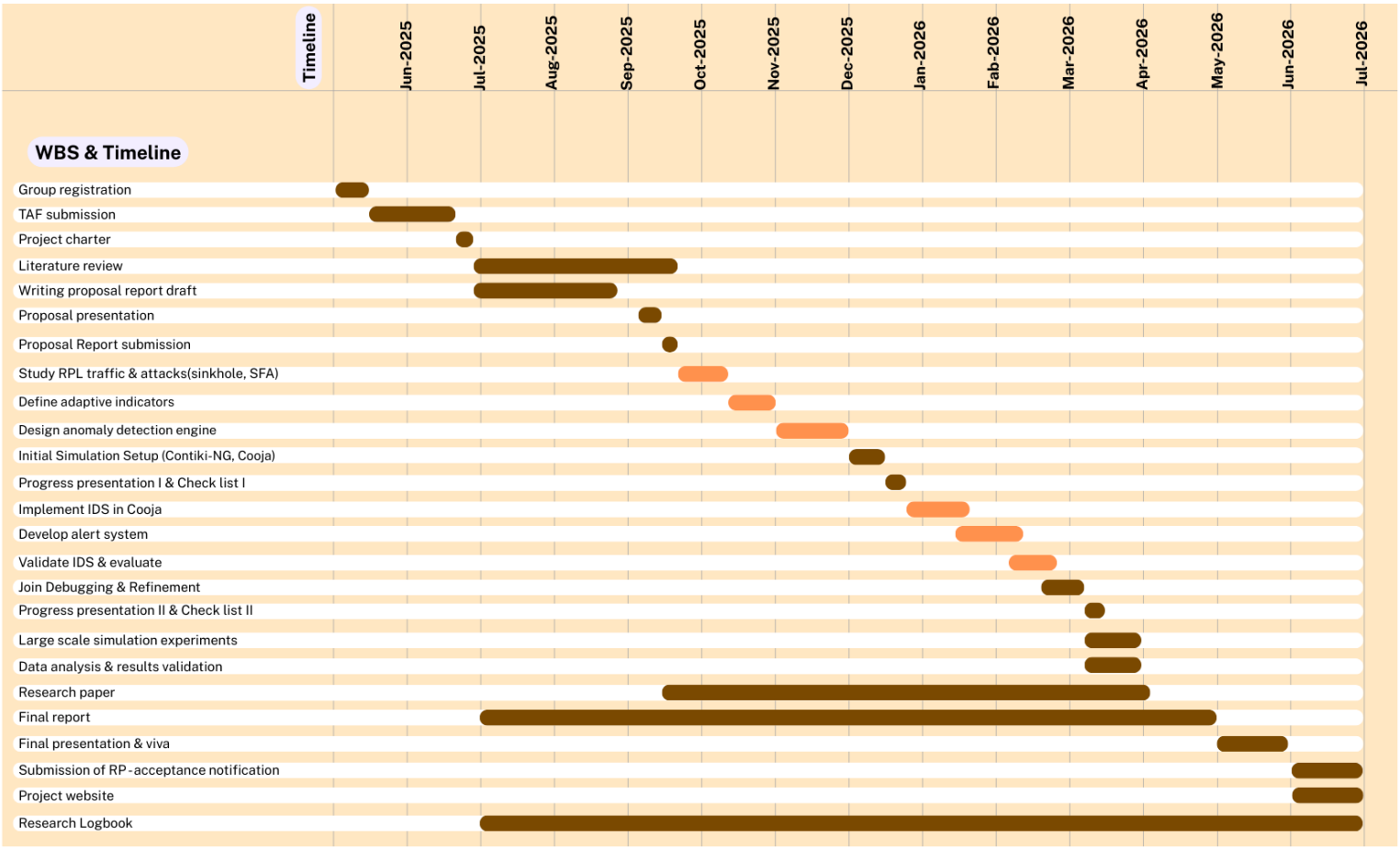
One more boundary has to with the littleness of the research. The intrusion detection system is designed to be able to detect sinkhole and selective forwarding attacks. Other RPL-related attacks like the Sybil attack, wormhole attack, or rank manipulation attacks are excluded on purposes of maintaining focus and feasibility. This does limit the coverage of attacks, but the coverage depth can be more impacting to the study then even more hinderance.

The study functions within a typical IoT environment's resource constraints, which limit the complexity of detection mechanisms that can be used. Therefore, the study steers clear of approaches that are heavy on computation, such as machine learning and deep learning models. This constraint influences the selection of a threshold-based detection strategy, yet stays true to the aim of developing a practical and lightweight IDS for constrained devices.

Additionally, the lack of assessment on the mobility of the network is restricted to certain mobility patterns that are supported by the simulation environment. These patterns are typical, but may not be ascertained to cover the full range of mobility activities that exist in the real world. However, these patterns are enough to evaluate the robustness of the proposed IDS within dynamic routing.

Driven by these constraints, the research does not lose either its feasibility or the academic rigor. Rather, it keeps the proposed work focused, lightweight and within the practical deployment scenario for RPL-based IoT networks.

10. Project plan or schedule



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12. Appendices

Appendix 1: Demonstration Video of the Proposed Intrusion Detection System

DEMO.mp4

In this demo, I present the first phase of my research, which focuses on implementing a lightweight, threshold-based intrusion detection system for RPL-based IoT networks. This phase aims to establish a low-overhead detection foundation suitable for constrained sensor nodes, without using computationally expensive learning models.

In the simulation, Node 1 functions as both the RPL root and UDP server, forming the RPL DODAG, while the remaining nodes act as clients that automatically join the topology and forward data hop-by-hop toward the root. The IDS runs locally on each node, allowing distributed monitoring of routing and forwarding behavior.

I first demonstrate the static network scenario, which serves as the baseline. Client nodes generate UDP traffic, and the IDS computes Packet Time Interval (PTI), Sequence Integrity Score (SIS), and Control-to-Data Flow Ratio (CDFR) using window-based measurements. Occasional PTI alerts appear due to normal delays, while SIS remains inactive, confirming correct baseline behavior and low severity scores.

Next, mobility is introduced to stress-test the system. As nodes move, routing instability increases control traffic and packet delays, which is reflected in PTI and CDFR logs. SIS remains largely unaffected, confirming that it correctly detects true forwarding anomalies rather than mobility effects. Severity scores increase only when multiple metrics exceed thresholds within the same observation window.

This phase validates the IDS foundation, including metric computation, threshold-based detection, and severity scoring under static and mobile conditions. In the next phase, explicit sinkhole and selective forwarding attacks will be introduced to evaluate individual and simultaneous attack detection performance.